Yi Zhen — Curriculum Vitae

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Job Experience & Roles

Senior Software Engineer **Ant Group**
Team Lead & Tech Architect *Mar. 2021 – Present*
I have been building AI systems for optimization of LLM performance, operator development since 2024. I lead a small team to create a brand new open source smart contract compiler framework named SIR on AntChainOpenLabs in 2023. I also launched a university-enterprise cooperation project with SYSU. Before that, I was the lead developer and project manager of Gödel query language

Software Engineer **Huawei Research**
Tech Lead, Development *Nov. 2018 – Mar. 2021*
I act as the tech lead to design and implement the lowcode platform. Developer of open ark compiler

Software Engineer **Citigroup**
Development, Continuous Integration *Apr. 2017 – Nov. 2018*
I worked on the anti-money laundering platform and the wire transfer system for the block trade agent

Miscellaneous

Publications

- X. Xie, G. Fan, X. Lin, A. Zhou, S. Li, X. Zheng, Y. Liang, Y. Zhang, N. Yu, H. Li, X. Chen, Y. Chen, **Y. Zhen**, D. Dong, X. Fu, J. Su, F. Pan, P. Luo, Y. Feng, R. Hu, H. Guo, J. Fan, X. Xiao, P. Di, "Principles and Practices of Large-Scale Code Analysis at Ant Group: A Data- and Logic-Oriented Approach," To appear in the IEEE/ACM International Conference on Software Engineering (**ICSE 2026**)
- Z. Li, J. Wu, Z. Wu, D. Tan, Z. Wei, Z. Jiang, **Y. Zhen**, Z. Zhang and Z. Zheng, "CCIHunter: Enhancing Smart Contract Code-Comment Inconsistencies Detection via Two-Stage Pre-training," To appear in the ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**)
- Z. Wu, J. Wu, H. Zhang, Z. Li, J. Chen, Z. Zheng, Q. Xia, G. Fan and **Y. Zhen**, "DAppFL: Just-in-Time Fault Localization for Decentralized Applications in Web3," To appear in the 33rd edition of the ACM SIGSOFT International Symposium on Software Testing and Analysis (**ISSTA 2024**)

Representative Projects

LLM Optimization of Training and Inference **tech lead**
operators library & LLM evaluation *Feb. 2024 – Present*
LLM deployment and training, model compression, LLM performance profiling, development and tuning of the operator library, evaluation and testing of LLM

Smart Intermediate Representation **co-founder**
smart contract compiler framework *Nov. 2022 – Jan. 2024*
A universal technical solution that can be adapted to any programming language to help the blockchain platform/developer make their smart contracts safer, high performance and multi-scenario compatible
Github Address `https://github.com/AntChainOpenLabs/Smart-Intermediate-Representation`

Gödel Query Language **project manager**

Query language and compiler with Datalog engine underneath *Jul. 2021 – Oct. 2022*

Gödel query language is a DSL for querying and data processing used by CodeFuse-Query. Gödel uses a similar syntax to Rust, providing strict type checking, convenient type inference, and user-friendly error messages

Github Address `https://github.com/codefuse-ai/CodeFuse-Query/tree/main/godel-script`

ERC721G: Anti-Theft ERC721 Solution **co-founder**

web3 smart contract *Jun. 2022 – Oct. 2022*

World's First Anti-Theft NFT Series and Solution: A universal technical solution to be adapted to any ERC721 standard smart contract to help NFT holders guard their property security jointly

Github Address `https://github.com/turtlecasedao/OpenERC721G`

OpenSea Address `https://opensea.io/collection/turtle-case-gang`

PPrinter: A generic derivable Haskell pretty printer **author**

Haskell Library - 3913 times download till 02/03/2026 *Jun. 2016 – Aug. 2016*

PPrinter is a Haskell library that supports automatic derivation of pretty printing functions on user defined arbitrary data types (the deriving mechanism supports the automatic generation of instances for functions)

Hackage Address `http://hackage.haskell.org/package/PPrinter-0.1.0`

International Student Supercomputer Challenge

Sun Yat-sen University ASC Student Supercomputer Challenge Team

Team member *2014*

- ASC14 required the team to wring the most HPC performance out of a 3000W power allowance
- Mastered the numerical methods, GPGPU heterogeneous computing and multiprocessor programming
- Optimized the SU^2 – a Stanford University developed open-source C++ code for PDE analysis and designed things that adhere to PDE constraints, and assisted in the HPL event

Professional Skills

Programming Language and Framework

Language: Triton/Tilelang, Rust, Solidity, TypeScript, Haskell, C/C++, Java, Python, Scala, Prolog, Coq

Framework: SGLang, vLLM, CUDA, MLIR/LLVM, ONNXRuntime, Hardhat, web3.js/ether.js, ASM

Awards

First Prize and Highest Linpack Award

The ASC Student Supercomputer Challenge (ASC14) *2014*

Set a new world record of HPL(Linpack) performance and won 10,000 CNY

Bronze Medal

The ACM-ICPC China Guangdong Provincial Programming Contest (GDCPC) *2014*

Two-time recipient of First Prize

The National Olympiad in Informatics in Provinces (NOIP) *2009&2008*

Education

University of Edinburgh **Edinburgh, U.K.**

Master of Science in Artificial Intelligence *Sep. 2015 – Nov. 2016*

Dissertation: Deriving Pretty-printing for Haskell, supervised by Prof. Philip Wadler

Sun Yat-sen University **Guangzhou, P.R.China**

Bachelor of Engineering in Software Engineering *Sep. 2011 – Jun. 2015*

Recommended for admission to SYSU and exempted from Gaokao because of well performance at NOIP